Predictive Modeling and Risk Scoring for Bank Customer Churn

A Machine-Learning Research Paper with Explainable Risk Scores and What-If Analysis

Unified Mentor Internship Project

Prepared by: Lokesh

12 October 2026

Abstract

Customer churn erodes revenue and customer lifetime value in retail banking, yet most retention effort is reactive and broad. This paper builds a predictive churn-intelligence workflow on a dataset of 10,000 bank customers in France, Germany and Spain, of whom 20.4% exited. After preprocessing, feature engineering and a stratified train-test split, four classifiers (logistic regression, decision tree, random forest and gradient boosting) were compared on accuracy, precision, recall, F1-score and ROC-AUC. Gradient Boosting was selected on cross-validated ROC-AUC and reached a test ROC-AUC of 0.869; at a tuned decision threshold of 0.37 it achieved a recall of 59.5% and a precision of 68.8%. Permutation importance and partial dependence analysis identify Age, NumOfProducts, Geography Germany as the leading churn drivers. Predicted probabilities are converted into low-, medium- and high-risk segments, and a what-if analysis shows how changes in engagement and product holding move an individual customer's risk. The paper closes with targeted retention recommendations and the specification of a Streamlit risk-scoring dashboard.

Keywords: customer churn; risk scoring; random forest; gradient boosting; explainable machine learning; retail banking; retention analytics

Executive Summary

Why it matters. About 20.4% of the customers in this portfolio left the bank. Each departure removes future revenue and cross-sell potential, and acquiring a new customer typically costs more than retaining an existing one. Today most retention action starts after the customer has already decided to leave.

What we did. We trained and compared four prediction models on 10,000 customer records and turned the best one into a churn-probability score between 0 and 1 for every customer, so that retention teams can act before a customer leaves.

What we found.

- The best model (Gradient Boosting) ranks customers well: its ROC-AUC on unseen customers is 0.87 (1.0 is perfect, 0.5 is guessing).
- The customers scored as high risk (probability of 50% or more) make up 12.8% of the customer base but account for 48.9% of all actual churners; their observed churn rate is 77.7%.
- The strongest drivers of churn are: Age, NumOfProducts, Geography Germany, Balance, EngagementProduct.
- Churn differs sharply between groups: by country (Germany highest at 32.4%), by activity status (inactive members 26.9% vs active 14.3%), and by number of products held.

What we recommend.

- Use the churn score to focus retention offers on the highest-risk segment instead of contacting everyone.
- Run re-engagement programmes for inactive members and review the product journey of customers holding an unusual number of products.
- Deploy the Streamlit dashboard so relationship managers can test what-if scenarios, and monitor and retrain the model regularly.

1. Introduction

1.1 Background and context

Customer churn directly affects customer lifetime value (CLV), revenue stability, cross-sell and upsell potential, and the long-term competitiveness of retail banks. Traditional churn analysis explains why churn happened after the fact. Modern banking instead needs to identify, early, the customers who are likely to leave, so that it can run proactive retention campaigns, personalised offers and targeted customer engagement. This project therefore builds a predictive churn-intelligence system that assigns a risk probability to each customer before they leave.

1.2 Problem statement

Despite holding rich customer-level data, banks often lack accurate churn prediction models, quantitative churn risk scores and explainable insight into churn drivers. As a result, retention actions are reactive rather than proactive, broad rather than targeted, and therefore inefficient and costly.

1.3 Objectives

Primary objectives:

- Predict customer churn with high accuracy.
- Generate churn probability scores.
- Identify the key churn drivers.

Secondary objectives:

- Reduce false positives in churn detection.
- Improve the interpretability of machine-learning models.
- Enable scenario-based churn risk analysis.

1.4 Structure of the paper

Section 2 describes the data, Section 3 the methodology, and Section 4 the exploratory analysis. Sections 5 and 6 report model performance and explainability. Section 7 turns predictions into risk segments and what-if scenarios, Section 8 specifies the Streamlit dashboard, and Sections 9 to 11 give recommendations, limitations and conclusions.

2. Dataset Description

The dataset contains one row per customer of a European retail bank with 10,000 records and 14 original columns. The target variable is Exited (1 = customer churned, 0 = customer retained). Table 1 lists the variables.

Table 1. *Dataset dictionary.*

Column	Description	Type
Year	Year of the record (single constant value; removed)	Numeric
CustomerId	Unique customer identifier (removed - non-informative)	ID
Surname	Customer surname (removed - non-informative)	Text
CreditScore	Customer creditworthiness	Numeric
Geography	France, Spain or Germany	Categorical
Gender	Male / Female	Categorical
Age	Customer age in years	Numeric
Tenure	Years with the bank	Numeric

Column	Description	Type
Balance	Account balance	Numeric
NumOfProducts	Number of bank products held	Numeric
HasCrCard	Credit card ownership (0/1)	Binary
IsActiveMember	Activity indicator (0/1)	Binary
EstimatedSalary	Estimated annual salary	Numeric
Exited	Churn indicator (target)	Binary

Data quality checks found 0 missing values and 0 duplicate customer rows after removing identifier columns. 20.4% of customers churned (2,037 of 10,000), so the classes are imbalanced; this is handled with a stratified split and decision-threshold tuning rather than by altering the data.

3. Methodology

3.1 Data preprocessing

- Missing values are imputed (median for numeric, mode for categorical); none were found in this dataset.
- Non-informative columns (RowNumber, CustomerId, Surname, and the constant Year column when present) are removed.
- Geography and Gender are one-hot encoded (France and Female act as reference categories).
- Numerical features are standardised for logistic regression, whose coefficients and optimiser are scale-sensitive; tree-based models are scale-invariant and use the raw values.

3.2 Feature engineering

Table 2. *Engineered features.*

Derived feature	Definition	Rationale
BalanceSalaryRatio	Balance / (EstimatedSalary + 1)	Share of income held with the bank
ProductDensity	NumOfProducts / (Tenure + 1)	Speed of product adoption
EngagementProduct	IsActiveMember x NumOfProducts	Engagement combined with product depth
AgeTenure	Age x Tenure	Life-stage and loyalty interaction
TenureAgeRatio	Tenure / Age	Share of adult life spent with the bank

3.3 Train-test strategy

The data are split 80/20 with stratification on Exited so that the churn rate is preserved (train 20.4% churn, 8,000 customers; test 20.3% churn, 2,000 customers). Five-fold stratified cross-validation on the training set is used for model selection and for choosing the decision threshold, so that the test set is used only once for final reporting.

3.4 Models

- **Logistic regression** - interpretability benchmark.
- **Decision tree** - transparent rule-based model.
- **Random forest** - bagged ensemble of trees that reduces variance.
- **Gradient boosting** - sequential ensemble that corrects the errors of earlier trees; typically the strongest tabular learner. XGBoost was not required for the brief and can be substituted without changing the pipeline.

Hyper-parameters are listed in Appendix A.

3.5 Evaluation metrics

Table 3. *Evaluation metrics.*

Metric	Purpose
Accuracy	Overall correctness
Precision	Controls false churn alarms
Recall	Captures actual churners
F1-score	Balances precision and recall
ROC-AUC	Model discrimination across all thresholds

3.6 Explainability

Feature importance is measured by permutation importance on the test set (the fall in ROC-AUC when one feature is randomly shuffled), and the shape of each effect is shown with partial dependence plots. SHAP values can be added in the accompanying notebook for per-customer attributions (Lundberg and Lee, 2017); permutation importance and partial dependence were used here because they are model-agnostic and need no additional dependencies.

3.7 Prediction output and risk scoring

For every customer the model returns (i) a churn probability between 0 and 1 and (ii) a binary churn flag obtained by comparing that probability with a threshold. The threshold is chosen to maximise the F1-score on out-of-fold training predictions. Probabilities are grouped into three risk segments: Low (below 25%), Medium (25-50%) and High (50% or above).

4. Exploratory Data Analysis

4.1 Target distribution

20.4% of 10,000 customers churned (Figure 1), a ratio of roughly 1 churner to 3.9 retained customers. Because of this imbalance accuracy alone is misleading - a model that predicts "retained" for everyone would already be 79.6% accurate - so precision, recall, F1 and ROC-AUC are reported alongside it.

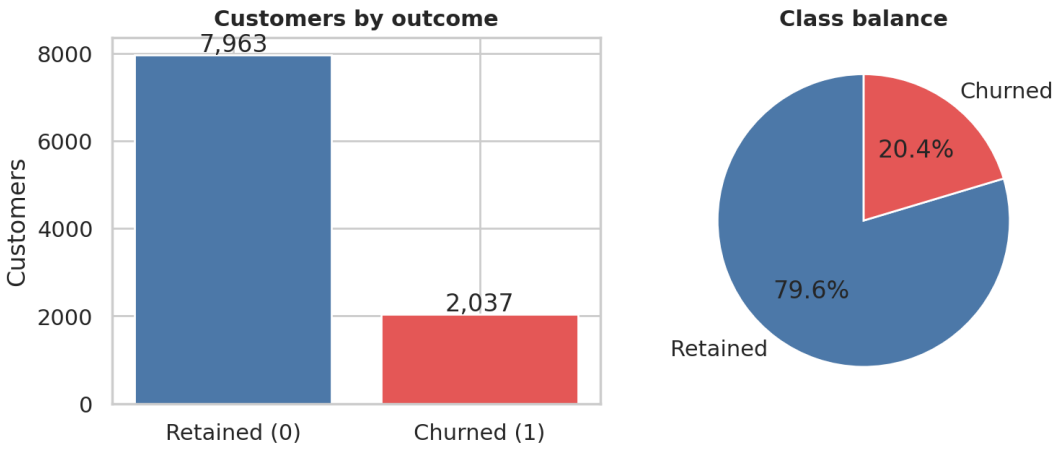


Figure 1. *Distribution of the target variable (Exited): customer counts and class balance.*

4.2 Age

Churned customers are on average 44.8 years old versus 37.4 for retained customers. The 51-60 age band has the highest churn rate (56.2%), while the 18-30 band has the lowest (7.5%) (Figure 2).

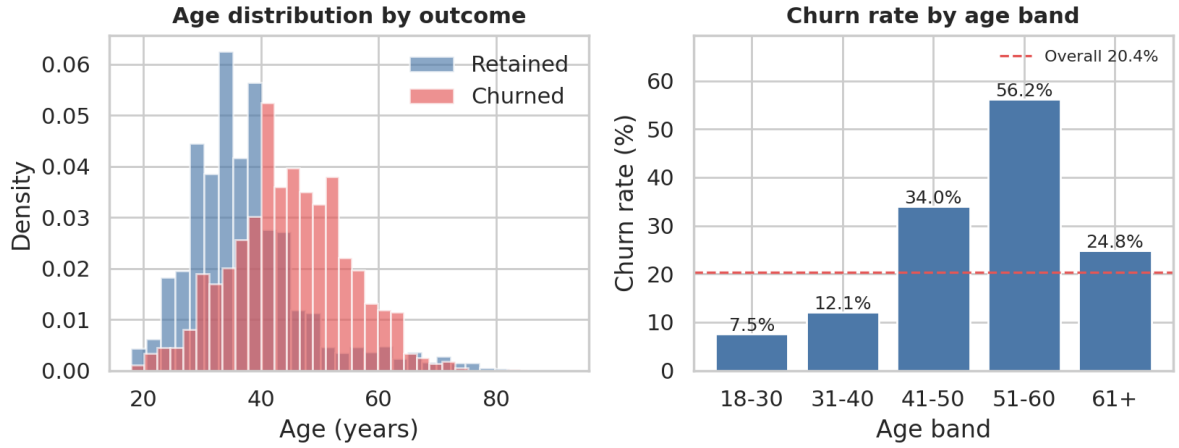


Figure 2. *Age profile of churned versus retained customers and churn rate by age band.*

4.3 Geography and gender

Churn differs by country: Germany is highest at 32.4% and France is lowest at 16.2% (overall 20.4%). By gender, Female customers churn more (25.1%) than Male customers (16.5%) (Figure 3).

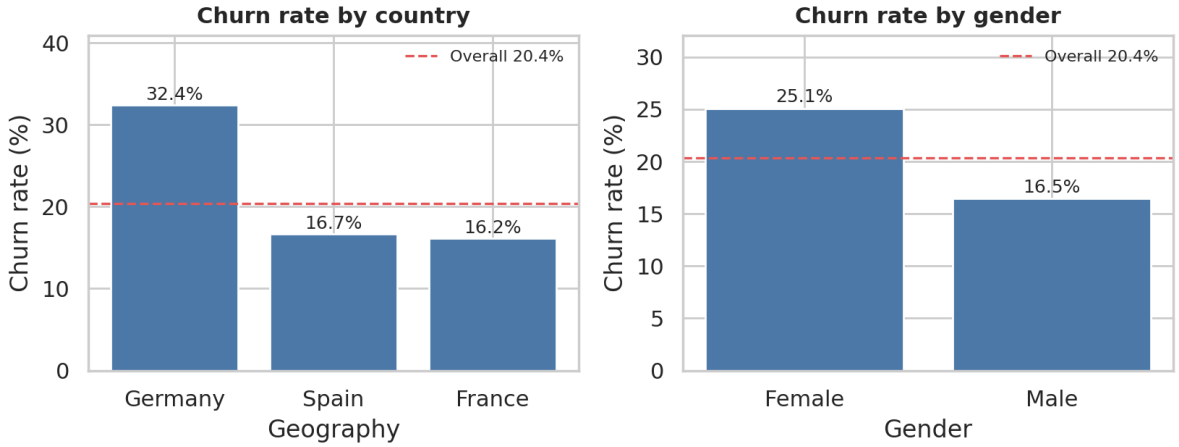


Figure 3. *Churn rate by country (left) and by gender (right); dashed line = overall churn rate.*

4.4 Product holding and engagement

Churn rate varies strongly with the number of products held, from 7.6% for customers with 2 product(s) to 100.0% for those with 4 (Figure 4; group sizes: 1: 5084, 2: 4590, 3: 266, 4: 60). Inactive members churn at 26.9% versus 14.3% for active members; engagement is therefore a clear marker of retention risk.

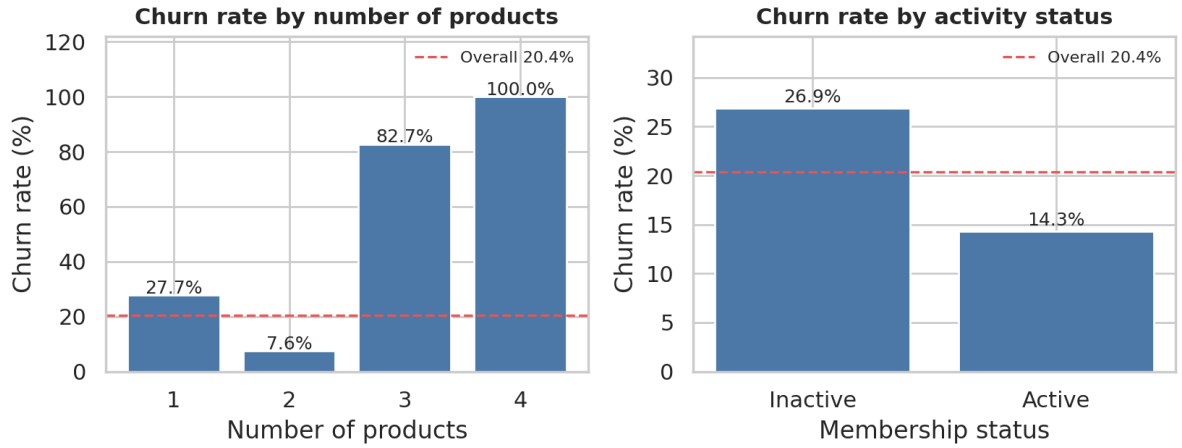


Figure 4. *Churn rate by number of bank products held (left) and by active-member status (right).*

4.5 Financial characteristics

Mean balance is 91,109 for churned and 72,745 for retained customers, and 24.5% of churned customers hold a zero balance versus 39.1% of retained customers. Mean credit score (645 vs 652) and estimated salary (101,466 vs 99,738) differ much less (Figure 5).

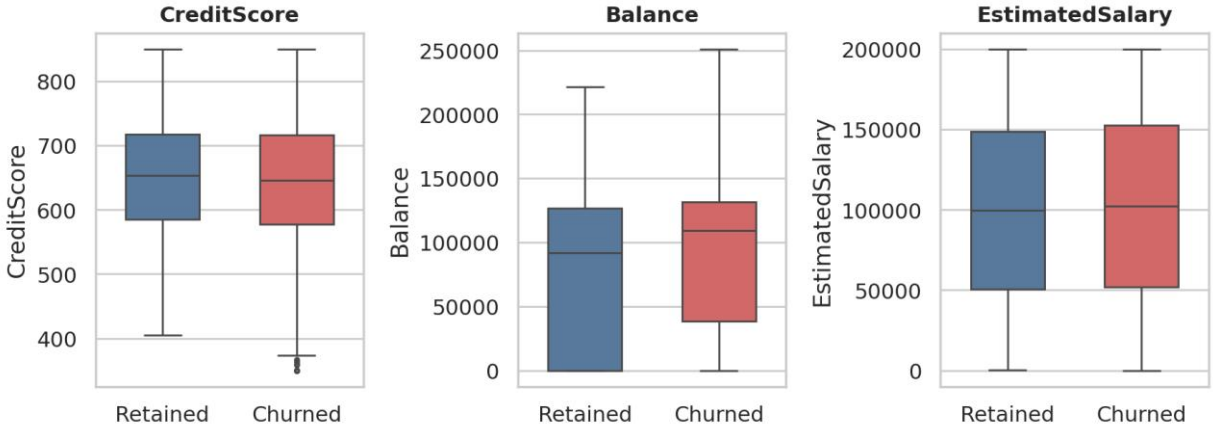


Figure 5. *Credit score, account balance and estimated salary for retained versus churned customers.*

4.6 Correlation structure

The features most correlated with churn are Age (r = +0.29); Geography Germany (r = +0.17); IsActiveMember (r = -0.16); EngagementProduct (r = -0.14) (Figure 6). Several engineered features are correlated with the raw variables they are built from, which is acceptable for tree-based models but is one reason the logistic regression is regularised.

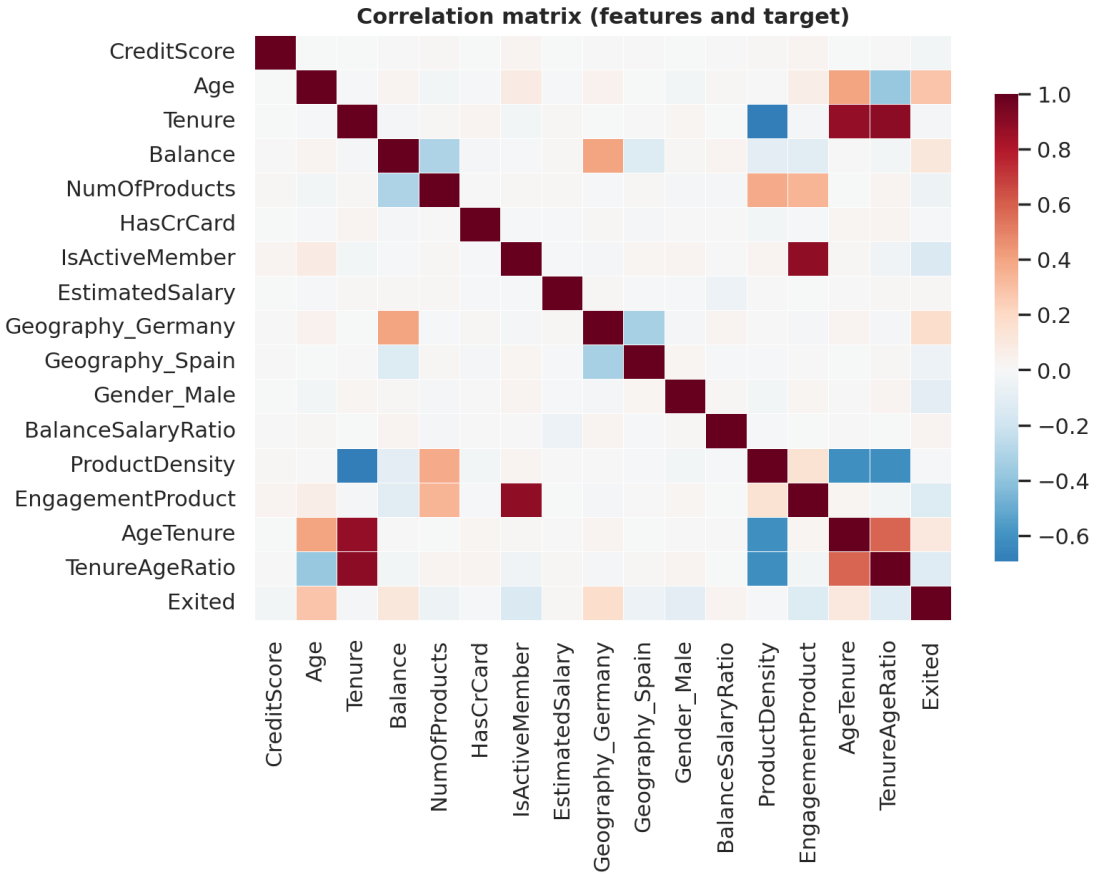


Figure 6. *Pearson correlation matrix of engineered features and the target (Exited).*

5. Model Performance

Table 4 compares the four models on the held-out test set at the default threshold of 0.50. Gradient Boosting was selected because it had the highest five-fold out-of-fold ROC-AUC on the training data (0.862).

Table 4. *Test-set performance at threshold 0.50 (CV AUC = five-fold out-of-fold ROC-AUC on the training set).*

Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV AUC
Logistic Regression	0.813	0.624	0.204	0.307	0.777	0.768
Decision Tree	0.861	0.788	0.430	0.556	0.847	0.833
Random Forest	0.868	0.812	0.457	0.585	0.863	0.856
Gradient Boosting	0.868	0.777	0.489	0.600	0.869	0.862

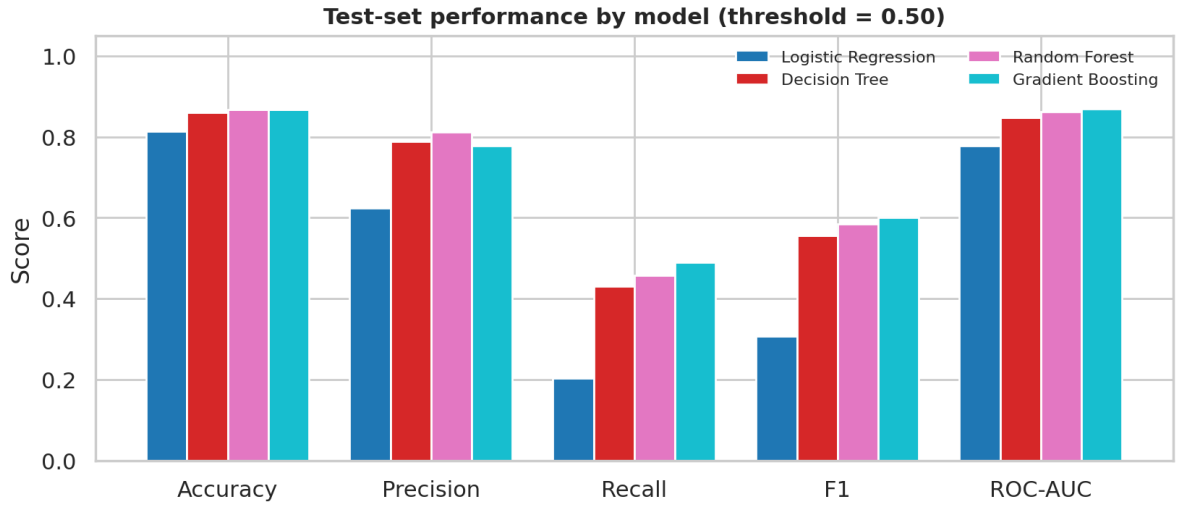


Figure 7. *Comparison of the four models on accuracy, precision, recall, F1-score and ROC-AUC (test set).*

By ROC-AUC the ranking is Gradient Boosting (0.869) > Random Forest (0.863) > Decision Tree (0.847) > Logistic Regression (0.777) (Figure 8). The selected model's ROC-AUC differs from that of the logistic-regression benchmark by +0.092, which indicates how much non-linear structure and interactions add on top of a linear model.

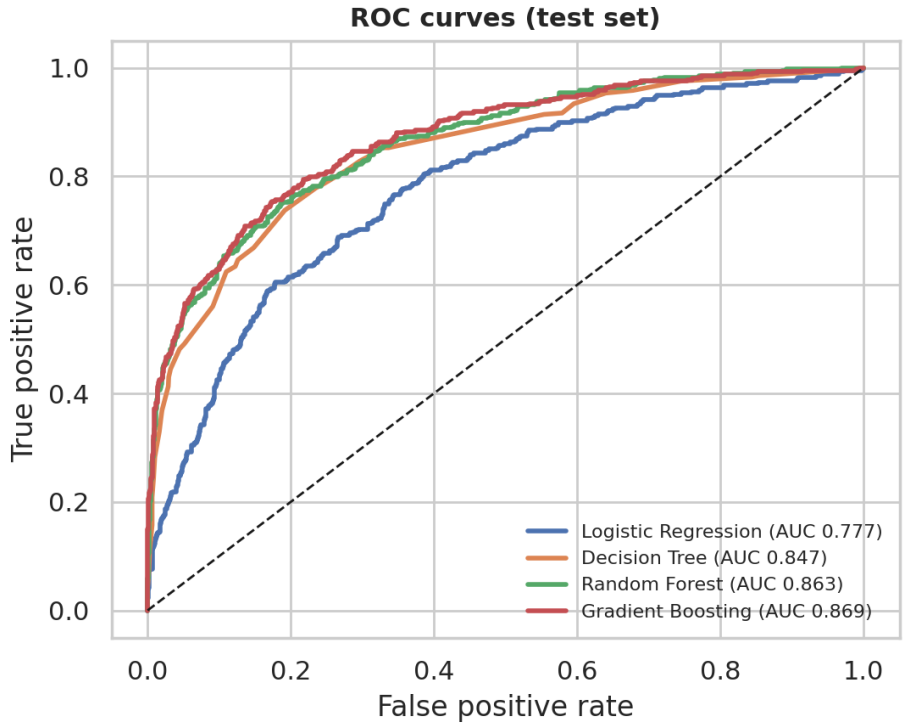


Figure 8. *Receiver operating characteristic curves for all models.*

5.1 Decision threshold and false positives

At the default threshold the selected model gives precision 77.7% and recall 48.9%. Because the classes are imbalanced, 0.50 is rarely the best operating point. Choosing the threshold that maximises F1 on out-of-fold training predictions gives 0.37, with test precision 68.8%, recall 59.5% and F1 0.638 (Table 5). The confusion matrix and the precision-recall trade-off are in Figure 9; a retention team that wants fewer false alarms can raise the threshold, while one that wants to capture more churners can lower it.

Table 5. *Selected model at the default and tuned decision thresholds.*

Operating point	Threshold	Accuracy	Precision	Recall	F1
Default	0.50	0.868	0.777	0.489	0.600
Tuned (max F1)	0.37	0.863	0.688	0.595	0.638

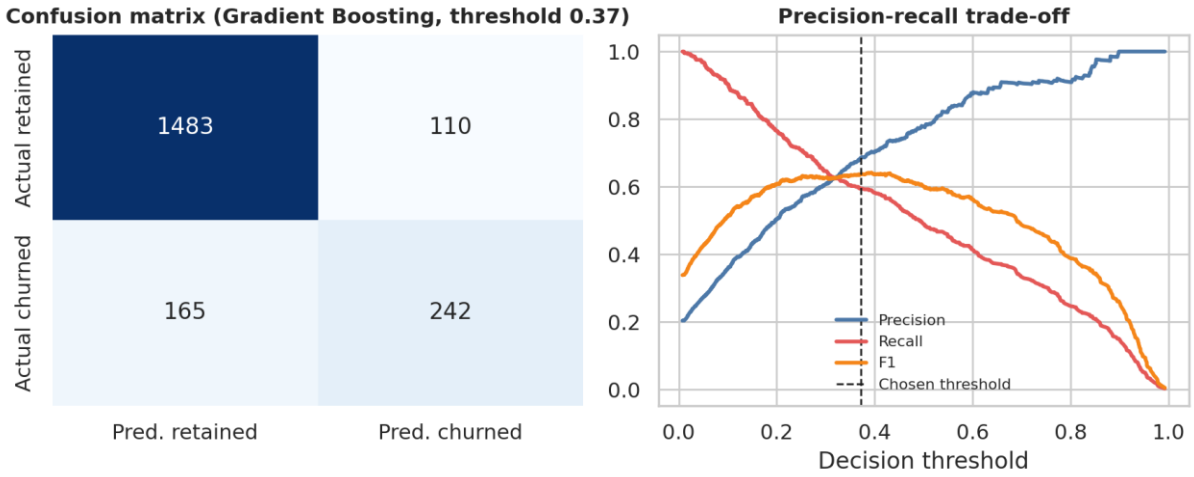


Figure 9. *Confusion matrix of the selected model at the tuned threshold (left) and the precision-recall-F1 trade-off across thresholds (right).*

6. Model Explainability

Figure 10 ranks features by how much shuffling them reduces ROC-AUC. The leading drivers are Age (0.115), NumOfProducts (0.106), Geography Germany (0.020), Balance (0.017), EngagementProduct (0.010). The top five features account for 91.1% of the total measured importance.

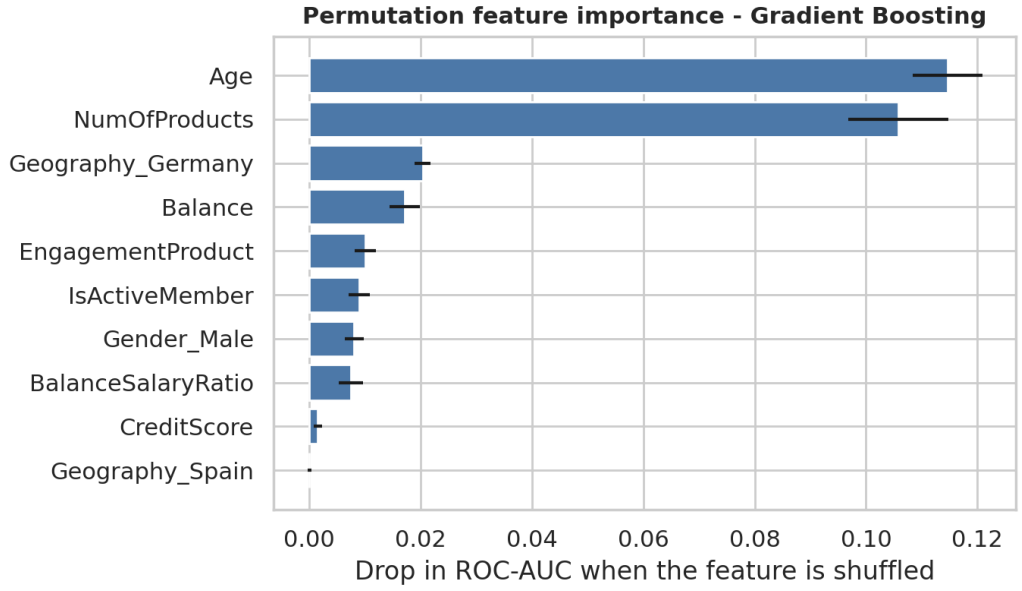


Figure 10. *Permutation feature importance of the selected model (top 10 features, test set).*

Figure 11 shows how the predicted churn probability changes as each of the four most influential non-binary features varies while the others are averaged out. These curves show the direction and shape of

each effect, including non-linear and threshold behaviour that a linear model cannot represent. Explainability of this kind is essential for regulatory compliance and for business trust in the model.

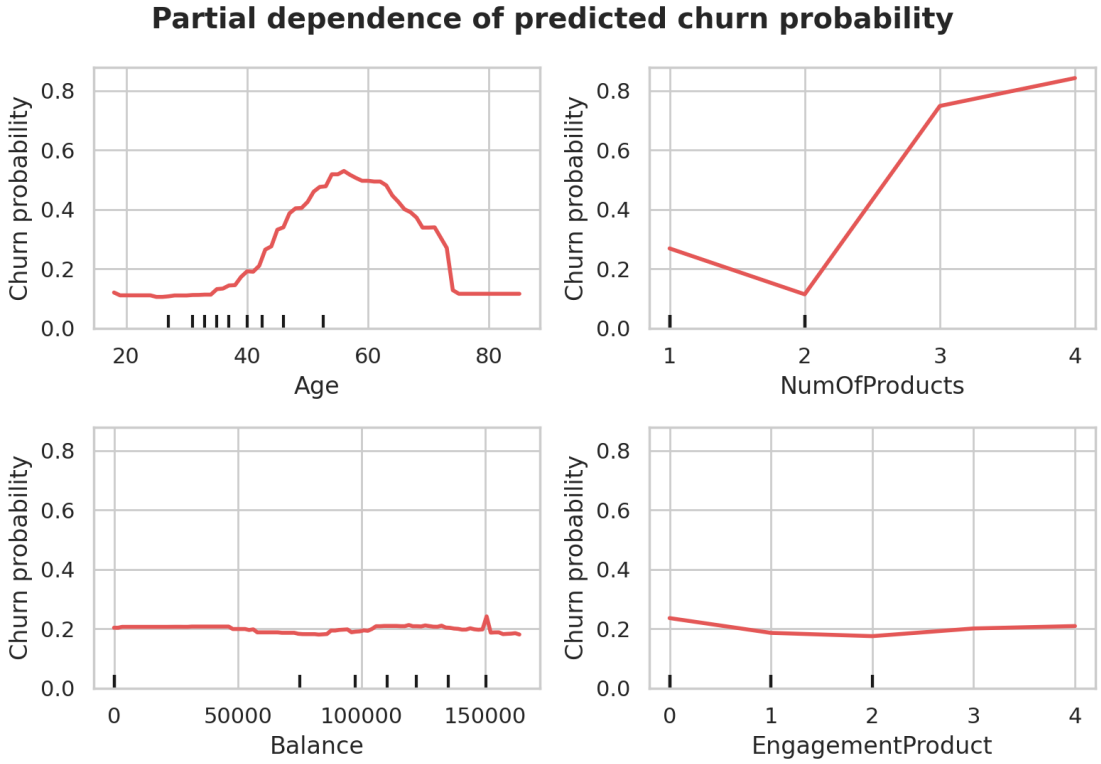


Figure 11. *Partial dependence plots for the four most influential non-binary features.*

7. Risk Scoring and Scenario Analysis

7.1 Probability distribution and risk segments

Figure 12 shows the distribution of predicted probabilities for retained and churned customers; the better the separation, the more useful the score. Table 6 groups customers into risk segments. The highest-risk segment contains 12.8% of customers but 48.9% of all churners, and the top 10% of customers by score churn at 86.5% - 4.3 times the portfolio average (lift).

Table 6. *Risk segments on the test set.*

Risk segment	Customers	Share of customers	Observed churn rate	Share of all churners
Low (<25%)	1488	74.4%	8.0%	29.2%
Medium (25-50%)	256	12.8%	34.8%	21.9%
High (>=50%)	256	12.8%	77.7%	48.9%

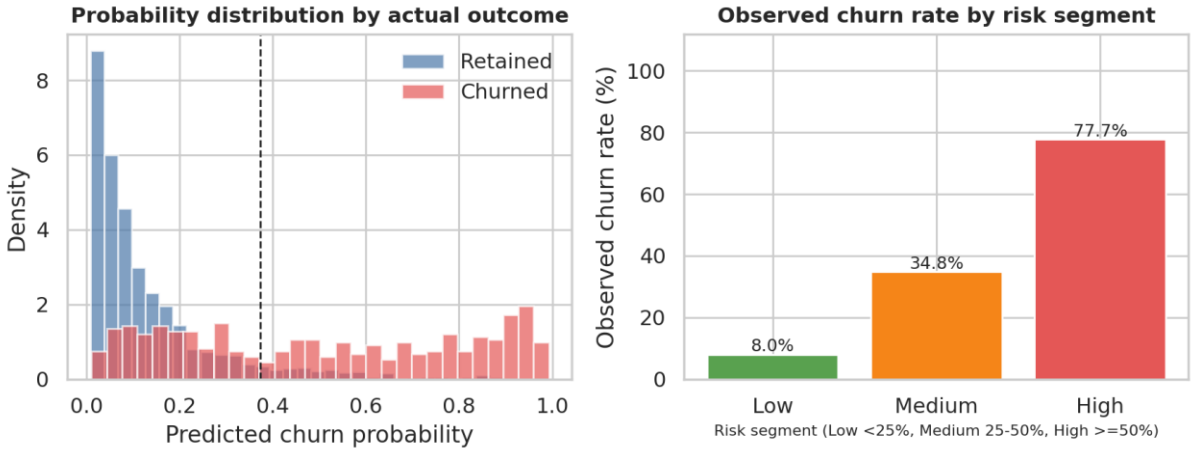


Figure 12. *Distribution of predicted churn probabilities (dashed line = decision threshold) and observed churn rate in each risk segment.*

7.2 What-if scenario analysis

To illustrate the scenario simulator, a baseline customer is defined with the median or most common value of each variable: France, Male, age 37, tenure 5 years, balance 97,199, 1 product(s), active member. Table 7 and Figure 13 show how the churn probability changes when one or more characteristics change.

Table 7. *What-if scenarios for the baseline customer (pp = percentage points).*

Scenario	Churn probability	Change vs baseline
Baseline (median customer, active member)	5.9%	-
Becomes an inactive member	9.5%	+3.6 pp
Ten years older	19.0%	+13.2 pp
Holds 2 products	3.5%	-2.3 pp
Holds 3 products	81.5%	+75.6 pp
Account balance doubled	15.1%	+9.2 pp
Located in Germany	15.9%	+10.1 pp
Inactive + ten years older + Germany	69.3%	+63.5 pp

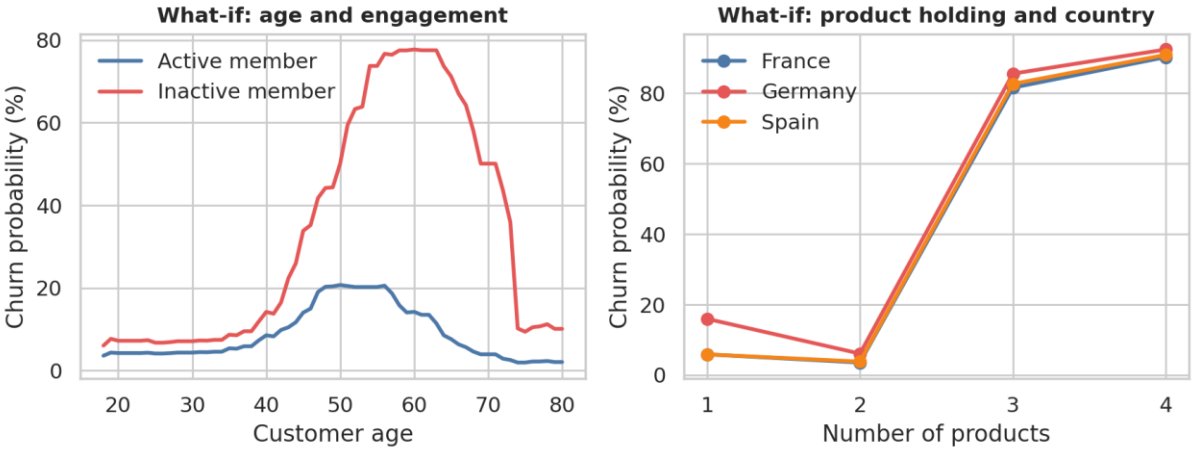


Figure 13. *What-if simulation for the baseline customer: churn probability as age, engagement, number of products and country change.*

8. Streamlit Dashboard Specification

The trained model is exposed through a Streamlit web application so that non-technical users can score customers and explore scenarios. The model and the feature-engineering function are saved once (joblib) and loaded with cached resources, so every interaction recomputes only the prediction.

Table 8. *Dashboard modules.*

Module	What the user sees / does
Customer churn risk calculator	Form with the customer's attributes; returns the churn probability, the binary flag and the risk segment.
Probability distribution	Histogram of portfolio scores with the current customer marked; segment sizes.
Feature importance dashboard	Ranked permutation importance and partial-dependence charts.
What-if scenario simulator	Sliders for age, activity, products and balance; probability updates live and the change from baseline is shown.

User capabilities are to input customer features, adjust engagement and product values, and observe how the churn probability changes.

9. Recommendations

1. **Target retention spend using the score.** The high-risk segment (12.8% of customers) contains 48.9% of churners. Prioritise personalised offers and relationship-manager calls there instead of broad campaigns.
2. **Re-engage inactive members.** Inactive members churn at 26.9% versus 14.3% for active members. Trigger re-activation journeys (usage nudges, product reviews, loyalty benefits) as soon as activity declines.
3. **Review the product journey.** Churn is 7.6% for customers with 2 product(s) and 100.0% for those with 4. Investigate pricing, onboarding and cross-sell practice in the highest-churn product groups before pushing further products.
4. **Investigate market-specific attrition.** Germany churn is 32.4% versus 16.2% in France. Examine competitor offers, fees and service quality in the highest-churn market.
5. **Build life-stage programmes.** The 51-60 age band churns at 56.2%. Design retirement-planning, wealth or family propositions suited to that stage.
6. **Check fairness and govern the model.** Because gender and country influence predictions, review outcomes for unintended bias, document the model, monitor score drift and retrain periodically.

10. Limitations and Future Work

- The data are a single snapshot without timestamps, so the model cannot learn how behaviour changes before churn; a time-based validation and transaction-level features would be a major improvement.
- Relationships are correlational. Retention actions should be validated with controlled experiments (A/B tests) before large-scale roll-out.
- Probabilities are not explicitly calibrated; calibration (e.g. isotonic regression) would sharpen the segment boundaries.
- Cost-sensitive thresholds that weigh the cost of an offer against the value of a retained customer were not modelled.

- Future work includes XGBoost/LightGBM, SHAP-based per-customer explanations, survival analysis for time-to-churn, and uplift modelling to identify who responds to an offer.

11. Conclusion

This study turned customer churn from a descriptive problem into a predictive, explainable risk-scoring workflow. Among four models, Gradient Boosting gave the best discrimination (test ROC-AUC 0.869) and, at a tuned threshold, identified 59.5% of churners with a precision of 68.8%. The analysis highlights Age, NumOfProducts, Geography Germany as the most influential drivers, and the what-if analysis shows how engagement and product holding change an individual customer's risk. Combined with the Streamlit dashboard and the recommendations above, the approach enables proactive, targeted and cost-efficient retention.

References

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5-32.

Friedman, J. H. (2001). Greedy function approximation: a gradient boosting machine. Annals of Statistics, 29(5), 1189-1232.

Lundberg, S. M., and Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems 30.

Pedregosa, F. et al. (2011). Scikit-learn: machine learning in Python. Journal of Machine Learning Research, 12, 2825-2830.

Hastie, T., Tibshirani, R., and Friedman, J. (2009). The Elements of Statistical Learning (2nd ed.). Springer.

Appendix A. Reproducibility

Table 9. *Model hyper-parameters.*

Model	Settings
Logistic Regression	StandardScaler + L2 logistic regression (max_iter=2000)
Decision Tree	max_depth=6, min_samples_leaf=30
Random Forest	400 trees, max_depth=10, min_samples_leaf=5
Gradient Boosting	300 trees, learning_rate=0.05, max_depth=3, subsample=0.8

Random seed 42; scikit-learn 1.8.0; 80/20 stratified split; five-fold stratified cross-validation on the training set. All numbers, tables and figures in this paper are produced by a single script (churn_paper_generator.py).